Collision-Site Control of Entanglement Structure in Small Qubit Networks

Research plan

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1 Problem and motivation

Consider a small network of qubits with fixed, always-on couplings along the edges of a graph G. From a product initial state the network can evolve into an entangled state whose structure is fixed by G: some subsets are entangled pairwise, some share genuine three- or four-qubit entanglement, others are not entangled at all. That structure need not match where entanglement is useful, for example at terminal nodes or on the qubits least exposed to noise.

Yunt *et al.* [1] showed in three-qubit networks that local collisions with fresh ancilla qubits at a chosen site route Bell-pair entanglement to different pairs, including pairs that never touch the ancilla. Here the object is the full multipartite structure, not only pairwise concurrence. Two questions are asked:

1. **Main question.** Can a local contact, coherent or collisional, reconfigure which remote subsets of a fixed connected graph are entangled, and with what entanglement depth?
2. **Ambitious question.** Can switching the contact site produce *stationary* genuine multipartite entanglement (GME) on qubits the ancilla never touches?

These involve different mechanisms, and the study must keep them apart.

2 Novelty, significance and two physical settings

What is known.

- Collision models describe open-system dynamics through repeated interactions with fresh ancillas [2].
- A shuttle qubit colliding sequentially with register qubits produces robust W-type entanglement among them [3].
- Engineered dissipation can prepare entangled and graph states as steady states [4, 5].
- Ising evolution from a uniformly x-polarized product state produces the graph state of the coupling graph [6]; a Pauli-Z measurement removes a vertex from a graph state [7].

- Repeated population projections (Zeno dephasing) steer spin-network dynamics [8].
- Decoherence-free subspaces and noiseless subsystems protect information through symmetry [9].
- Bell-pair routing by collision site in three-qubit chains and triangles [1].

Neither local production of entanglement nor the elimination of a strongly driven vertex is new by itself.

What is new.

1. A systematic, predictive relation between contact site, graph topology, ancilla preparation and the full multipartite entanglement structure: pairs, triples, the whole network, and separability with respect to every partition.
2. If verified in general, a switching protocol in which a first contact selects and a second contact stabilizes GME on uncontacted qubits.
3. A clean separation between coherent control by the contact and effects of fresh-ancilla dissipation.

Significance. Entanglement is consumed locally in quantum networks, but qubits are coupled globally and exposed to noise unevenly. A single-site control that reshapes or stabilizes the global entanglement structure needs no addressing of the target qubits.

Two physical settings.

- **Setting I: restructuring of naturally generated entanglement.** An XX network from $|0\cdots0\rangle$ creates graph-state-type entanglement on its own. A ZZ contact acts on it as a local Z field.
- **Setting II: collision-driven preparation and site switching.** An exchange network from the vacuum is inert. Population-inverted fresh ancillas pump excitations into it, and a sequence of contacts at different sites can select and stabilize entangled states.

Setting I asks how collisions *redistribute* existing entanglement; Setting II asks how they *generate and stabilize* it. Report the two separately.

3 Model

3.1 Network and collision map

Qubits $1,\dots,N$ sit on the vertices of a connected simple graph $G=(V,E)$. With $\hbar=1$,

$$H_G = J\sum_{(i,j)\in E} h_{ij}, \qquad h_{ij}\in\left\{X_iX_j,\ Z_iZ_j,\ \tfrac{1}{2}(\sigma_i^+\sigma_j^- + \sigma_i^-\sigma_j^+)\right\}, \qquad \sigma^\pm = \tfrac{1}{2}(X\pm iY). \tag{1}$$

A fresh ancilla a, prepared in η, couples to one network site v (the *contact site*) through $g\,h_{av}$ for a time τ, and is then discarded:

$$\mathcal{E}_{v,\eta}(\rho) = \mathrm{Tr}_a\left[U_v\,(\eta\otimes\rho)\,U_v^\dagger\right], \qquad U_v = e^{-i(H_G+g\,h_{av})\tau}. \tag{2}$$

The dimensionless controls are $J\tau$, g/J, the collision number n, and η. Do not confuse this fresh-ancilla channel with coherent evolution of one retained ancilla, the second protocol of [1].

Normalization. $\|XX\| = \|ZZ\| = 1$, but $\|\frac{1}{2}(\sigma^+\sigma^- + \text{h.c.})\| = \frac{1}{2}$. Do not compare coupling families at equal J or g without stating this.

3.2 The collision as a tunable control

For $h_{av} = \sigma_a^\alpha \sigma_v^\alpha$ ($\alpha = x, z$), the ancilla operator σ_a^α commutes with U_v. Expanding η in its eigenbasis gives the exact random-unitary form

$$\mathcal{E}_{v,\eta}(\rho) = p_+ \, U_+ \rho \, U_+^\dagger + p_- \, U_- \rho \, U_-^\dagger, \qquad U_\pm = e^{-i(H_G \pm g\,\sigma_v^\alpha)\tau}, \qquad p_\pm = \tfrac{1}{2}\left(1 \pm \langle \sigma_a^\alpha \rangle_\eta\right). \tag{3}$$

- $p_- = 0$ (ancilla eigenstate of σ^α): pure coherent control by a local field.
- $p_- = \frac{1}{2}$ (unpolarized ancilla; for a ZZ contact this includes $|+\rangle$): control plus a random field sign at each collision, which dephases.

Coherences of η in that eigenbasis have no effect. The ancilla *polarization* is the control parameter. For exchange contacts no ancilla Pauli is conserved and the full η matters.

3.3 Which coupling combinations act from the vacuum

Before any entanglement calculation, check which (network, contact) pairs can produce entanglement from $|0\cdots0\rangle$. Conserved quantities settle most cases:

Network	Contact	Conserved (network side)	Outcome from $\|0\cdots0\rangle$	Use
XX	ZZ	X_j for all $j \neq v$; $\Pi_z = \prod_i Z_i$	Weighted graph states, reshaped by a $\pm gZ_v$ field	Setting I
EX	EX	Joint excitation number	Network inert without collisions; pumping by ancilla $\|1\rangle$; full η matters	Setting II
EX	XX	Ancilla σ^x	Network inert without collisions; contact acts as a $\pm gX_v$ drive that injects excitations; strongly entangling	Setting II, coherent-drive comparison
XX	EX	Joint parity of network and ancilla	Graph states plus non-unital pumping at v	Optional
XX	XX	All X_j	Free evolution followed by a local channel on v; subsets without v unchanged	No
ZZ	XX or EX	Z_j for all $j \neq v$	Sites $j \neq v$ stay in $\|0\rangle$; product state	No
ZZ	ZZ	All Z_j	Nothing evolves	No
EX	ZZ	Network excitation number	Vacuum is stationary	No

The (EX, XX) pair is the configuration of Appendix B of [1]. With an X-eigenstate ancilla it is a coherent drive; with an unpolarized ancilla, a random drive.

Symmetry consequence for targets. In Setting I, Π_z is conserved. For odd N the z-basis GHZ state and the W state lie outside the reachable sector, so their fidelities can never certify GME there. Use x-basis GHZ states and graph states as targets.

Other initial states. ZZ and exchange networks do not entangle $|0\cdots0\rangle$ on their own. To study redistribution in those networks, prepare a suitable product superposition (e.g. $|+\rangle^{\otimes N}$) or excitation pattern. This is a separate study from Settings I and II.

3.4 Protocols

To compare contact sites on the same footing, fix the state before contact:

$$\rho_*(t_0) = e^{-iH_G t_0}\rho_0\, e^{iH_G t_0}, \qquad \rho_v(n;t_0) = \mathcal{E}^n_{v,\eta}\big(\rho_*(t_0)\big), \qquad \rho_{\text{free}}(n;t_0) = e^{-iH_G n\tau}\rho_*(t_0)\, e^{iH_G n\tau}. \tag{4}$$

Use $t_0 = 0$ for direct comparison with [1], and at least one t_0 fixed in advance at which the isolated network is entangled (for Setting I, $t_0 = \pi/4J$, where it is a graph state). The two-stage switching schedule is

$$\rho_{A\to B}(n_1,n_2) = \mathcal{E}^{n_2}_{B,\eta_2} \circ \mathcal{E}^{n_1}_{A,\eta_1}(\rho_0). \tag{5}$$

Compare it with A-only, B-only, the reversed order $B \to A$, and free continuations.

4 Entanglement toolbox

This section defines every quantity to be computed. Code is in Appendix A.

4.1 Reduced states, subsets and cuts

For $S \subseteq V$ with $|S| = k$, the reduced state is $\rho_S = \mathrm{Tr}_{V\setminus S}\,\rho$. There are $\binom{N}{k}$ subsets of size k; compute all of them, whether or not the induced subgraph $G[S]$ is connected. A k-qubit state has $2^{k-1}-1$ distinct bipartitions (cuts): 1 for a pair, 3 for a triple, 7 for four qubits. Fix the qubit ordering once (qubit 1 is the leftmost tensor factor) and never change it.

4.2 Partitions of four qubits

A *partition* of $V = \{1,2,3,4\}$ splits it into non-empty, disjoint blocks. There are 15 (the Bell number B_4):

Type	Number	Partitions
1234 (one block)	1	{1234}
1|3	4	1|234, 2|134, 3|124, 4|123
2|2	3	12|34, 13|24, 14|23
2|1|1	6	12|3|4, 13|2|4, 14|2|3, 23|1|4, 24|1|3, 34|1|2
1|1|1|1	1	1|2|3|4

The 7 two-block partitions are the cuts. Generate all partitions in code (Appendix A) and check that the routine returns 15.

Separability with respect to a partition. A state is P-*separable* for $P = \{B_1, \ldots, B_m\}$ if

$$\rho = \sum_k p_k \, \rho_{B_1}^{(k)} \otimes \cdots \otimes \rho_{B_m}^{(k)}, \qquad p_k \geq 0. \tag{6}$$

Entanglement may exist *within* each block but not *between* blocks. For example, $|\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34}$ is 12|34-separable but not 13|24-separable.

4.3 Entanglement between the parts of a cut: negativity

For a cut $A|B$ take the partial transpose on A and compute

$$\mathcal{N}_{A|B}(\rho) = \frac{\|\rho^{T_A}\|_1 - 1}{2} = \sum_{\lambda_i < 0} |\lambda_i(\rho^{T_A})|. \tag{7}$$

- $\mathcal{N}_{A|B} > 0$: entangled across the cut. This is a proof [10].
- $\mathcal{N}_{A|B} = 0$ (PPT): separable across the cut only for $2 \otimes 2$ and $2 \otimes 3$ systems [10], i.e. only for a pair of qubits. A 1|2 cut of three qubits is $2 \otimes 4$, and four-qubit cuts are $2 \otimes 8$ or $4 \otimes 4$; there PPT entangled states exist and a zero is inconclusive.

Compute all 7 cut negativities of the four-qubit state at every step.

4.4 Entanglement within a block

The entanglement inside a block B is a property of ρ_B:

- $|B| = 2$: Wootters concurrence [11]. Compute $\tilde{\rho} = (Y \otimes Y)\rho^*(Y \otimes Y)$ and the eigenvalues $\mu_i \geq 0$ of the Hermitian matrix $\sqrt{\rho}\,\tilde{\rho}\,\sqrt{\rho}$; with $\lambda_i = \sqrt{\mu_i}$ in decreasing order, $C = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4)$. Clip only round-off negatives ($|\mu_i| < 10^{-12}$) and flag anything larger. $C > 0$ if and only if the pair is entangled.
- $|B| = 3$: the three cut negativities of ρ_B and its GMN.
- $|B| = 4$: the full analysis of this section.

Reduced and global entanglement differ: a GHZ state has all $C_{ij} = 0$ yet is GME, while $|\Phi^+\rangle_{12}|0\rangle_3$ has $C_{12} = 1$ but no GME.

4.5 Genuine multipartite entanglement and GMN

A state is *biseparable* if it is a mixture of states each separable across *some* cut, possibly different cuts in different terms:

$$\rho = \sum_M p_M \, \rho_M, \qquad \rho_M \text{ separable across cut } M. \tag{8}$$

A state that is not biseparable is GME. Positive negativity across every cut does *not* imply GME for mixed states. For a *pure* state, $\mathrm{Tr}\rho_A^2 < 1$ for every cut $A|B$ is exactly equivalent to GME.

GMN. Replace "separable across M" by the larger set "PPT across M". The genuine multipartite negativity [12] is the value of a semidefinite program (SDP):

$$\mathrm{GMN}(\rho) = -\min_W \mathrm{Tr}(W\rho) \quad \text{s.t.} \quad W = P_M + Q_M^{T_M}, \;\; 0 \leq P_M \leq \mathbb{1}, \;\; 0 \leq Q_M \leq \mathbb{1} \quad \text{for every cut } M. \tag{9}$$

Biseparable states are PPT mixtures, so GMN > 0 proves GME. For three qubits it takes about one second; for four qubits use MOSEK or a real-valued reformulation.

A useful bound. For any feasible W and any cut M, $\mathrm{Tr}(P_M\rho) \geq 0$ and $\mathrm{Tr}(Q_M\rho^{T_M}) \geq -\mathcal{N}_M(\rho)$, because $0 \leq Q_M \leq \mathbb{1}$. Hence

$$\mathrm{GMN}(\rho) \leq \min_M \mathcal{N}_M(\rho). \tag{10}$$

If every cut negativity is below a tolerance δ, then GMN $< \delta$ rigorously. Use this to skip the SDP, but record the result as "GMN $< \delta$", not as an exact zero. Keep δ (10^{-9}) well below the certification threshold ϵ_{cert}.

Validation values (computed with the code of Appendix A).

State	Expected value
$\lvert\mathrm{GHZ}_3\rangle$	GMN $= 0.500$
$\lvert W_3\rangle$	GMN $= 0.443$
$\lvert\Phi^+\rangle \otimes \lvert 0\rangle$	GMN $= 0$; $C_{12} = 1$
$p\lvert\mathrm{GHZ}_3\rangle\langle\mathrm{GHZ}_3\rvert + (1-p)\mathbb{1}/8$	GMN $= 0$ for $p \leq 3/7$; positive above (0.010 at $p = 0.44$)
$p\lvert W_3\rangle\langle W_3\rvert + (1-p)\mathbb{1}/8$	GMN $= 0.088$ at $p = 0.60$
Werner: $p\lvert\Psi^-\rangle\langle\Psi^-\rvert + (1-p)\mathbb{1}/4$	$C = \max\left(0, \frac{3p-1}{2}\right)$

The GHZ threshold $p = 3/7$ is the exact GME boundary for this family [13]. Reproduce every row before using the code on network states.

4.6 Fidelity witnesses

For a pure target $\lvert\psi\rangle$, let

$$\alpha(\psi) = \max_{\text{cuts } A|B} \lambda_{\max}(\mathrm{Tr}_B \lvert\psi\rangle\langle\psi\rvert)\,, \tag{11}$$

the largest squared Schmidt coefficient over all cuts. Then $\langle\psi|\rho|\psi\rangle > \alpha(\psi)$ proves GME [14]. Values: $\alpha = \frac{1}{2}$ for GHZ states and for graph states of connected graphs; $\alpha = (k-1)/k$ for W_k. Compute α numerically for other targets. Optimizing the target over local unitaries leaves α unchanged.

Witnesses are weaker than GMN: the noisy W state at $p = 0.60$ has $F_W = 0.65 < \frac{2}{3}$, so the witness fails, while GMN $= 0.088$. A failed witness is inconclusive. Witnesses are still useful as an independent check and when GMN is expensive.

4.7 What to do when GMN is zero

A zero GMN means only that ρ is a PPT mixture. It does not mean the state is biseparable. Proceed as follows:

1. **Check the number.** Read the solver status and primal/dual residuals. Treat values below $\epsilon_{\mathrm{cert}} = \max(10 \times \text{residual}, 10^{-6})$ as uncertified. Re-solve near-threshold cases at tighter tolerance and with a second solver.
2. **If ρ_S is pure** (purity 1 within 10^{-12}), test $\mathrm{Tr}\rho_A^2 < 1$ on every cut. This decides GME exactly.
3. **If ρ_S is mixed**, record the full cut-negativity profile and all pair concurrences.
4. **Try fidelity witnesses** with targets inside the reachable symmetry sector: graph states of G and its subgraphs, x-basis GHZ states, and in Setting II W- and Dicke-type states.

5. **If the physical claim depends on it**, try stronger partition-aware SDP relaxations [12, 15] or construct an explicit biseparable decomposition. Only a valid decomposition establishes biseparability; feasibility of a relaxation does not.

Report each subset with one status:

- GME certified;
- biseparability established (explicit decomposition);
- other entanglement certified, GME unresolved;
- no entanglement certified.

4.8 Which partitions are excluded

For each of the 15 partitions P:

- If ρ is P-separable, it is separable across every cut $A|B$ that does not split a block of P (i.e. A is a union of blocks).
- Hence, if *any* block-respecting cut has $\mathcal{N}_{A|B} > 0$, then P is excluded.
- If all block-respecting cuts are PPT, P is *not excluded*. This is consistent with P-separability but does not establish it.

Example: for $|\Phi^+\rangle_{12}|\Phi^+\rangle_{34}$, the partition 12|34 is not excluded, while 13|24, 1|234 and 1|2|3|4 are excluded. Stronger criteria for the exact partition structure are in [15].

4.9 Entanglement depth

A state is *k-producible* if it is a mixture of products of pure states on blocks of at most k qubits. The depth D is the smallest such k. Only lower bounds are computed:

- **From subsets.** If some ρ_S is GME, then $D \geq |S|$. (If the global state were k-producible with $k < |S|$, every term restricted to S would be biseparable on S.)
- **From quantum Fisher information** [16]. For $J_{\mathbf{n}} = \frac{1}{2}\sum_i \mathbf{n}_i \cdot \boldsymbol{\sigma}_i$ with unit vectors $\mathbf{n}_i$, every k-producible state obeys $F_Q[\rho, J_{\mathbf{n}}] \leq sk^2 + r^2$, with $N = sk + r$, $0 \leq r < k$. For $N = 4$ the bounds are 4, 8, 10 for $k = 1, 2, 3$; violation certifies $D \geq k + 1$. Compute

$$F_Q[\rho, J] = 2 \sum_{i,j:\, \lambda_i + \lambda_j > 0} \frac{(\lambda_i - \lambda_j)^2}{\lambda_i + \lambda_j} |\langle i|J|j\rangle|^2 \tag{12}$$

  from $\rho = \sum_i \lambda_i |i\rangle\langle i|$, and maximize over the directions $\mathbf{n}_i$.

A positive triple GMN with zero four-qubit GMN means $D \geq 3$, not $D = 3$.

4.10 Reporting classes

Summarize each state with one label built only from certified quantities:

Label	Certificate
4-qubit GME	$\mathrm{GMN}_{1234} > \epsilon_{\mathrm{cert}}$, or a fidelity witness above α
$D \geq 3$, 4-qubit GME unresolved	Some triple with GMN $> \epsilon_{\mathrm{cert}}$; no 4-qubit certificate
$D \geq 2$, triples unresolved	Some $C_{ij} > 0$ or some NPT cut; no triple certified
No entanglement certified	All cut negativities and certificates zero

Add, as descriptive information, which of the 15 partitions are excluded (Sec. 4.8). Phrases such as "consistent with 1|3" describe PPT patterns and never replace a decomposition.

5 Interpretation tools

Each tool predicts something before any data are taken. Each benchmark result should be labelled as matching or breaking these predictions.

5.1 Contact-site symmetry

Let $\mathrm{Aut}(G)_v$ be the graph automorphisms that fix the contact site v. For each $\sigma \in \mathrm{Aut}(G)_v$ the qubit permutation P_σ commutes with the map (2). If the initial state and the ancilla schedule share the symmetry,

$$P_\sigma\, \rho(n)\, P_\sigma^\dagger = \rho(n) \qquad \forall\, \sigma \in \mathrm{Aut}(G)_v,\ \forall n. \tag{13}$$

Subsets related by such a σ then carry identical entanglement. This is exact; use it to check the code and to reduce the data.

Graph, contact	$\mathrm{Aut}(G)_v$	Forced equalities
Star, hub 1	S_3 on $\{2,3,4\}$	All leaf pairs equal; all hub–leaf pairs equal; triples $\{1,a,b\}$ equal
Star, leaf 2	swap $3 \leftrightarrow 4$	$(1,3)=(1,4)$; $(2,3)=(2,4)$; $\{1,2,3\}=\{1,2,4\}$
C_4, site 1	swap $2 \leftrightarrow 4$	$(1,2)=(1,4)$; $(2,3)=(3,4)$; $\{1,2,3\}=\{1,3,4\}$
Paw, vertex 3 or tail 4	swap $1 \leftrightarrow 2$	$(1,3)=(2,3)$; $(1,4)=(2,4)$; $\{1,3,4\}=\{2,3,4\}$
Diamond, vertex 2	swap $1 \leftrightarrow 3$	$(1,2)=(2,3)$; $(1,4)=(3,4)$; $\{1,2,4\}=\{2,3,4\}$
Diamond, vertex 1	swap $2 \leftrightarrow 4$	$(1,2)=(1,4)$; $(2,3)=(3,4)$; $\{1,2,3\}=\{1,3,4\}$
K_4, site 1	S_3 on $\{2,3,4\}$	As for the star hub
P_4 (end or middle), paw triangle vertex	trivial	None

(Graphs and labels are defined in Sec. 7.1.)

5.2 Symmetric pure states

A permutation-symmetric pure state is either fully separable or GME [17]. Prediction: when a contacted hub decouples (star or K_4, $g \gg J$, $p_- = 0$), the remaining three qubits are close to pure and symmetric, so they should show either no entanglement or GME, but no pairwise-only structure. A leaf contact breaks the symmetry and allows pair structures. For mixed states this is a hypothesis, not a theorem.

5.3 Graph-state classes

In Setting I without collisions, each factor $e^{-iJtX_iX_j}$ acts as a controlled-phase gate at $Jt = \pi/4$, so the network reaches the x-basis graph state of G [6]. Under local complementation the six connected four-vertex graphs fall into two local-unitary classes [7]:

- GHZ class: star, K_4;

- linear-cluster class: P_4, C_4, paw, diamond.

GHZ-class states have separable pair reductions. This predicts the no-collision baseline at $Jt = \pi/4$.

5.4 Fast/slow effective Hamiltonian (Setting I, $g \gg J$)

For a fixed sign $s = \pm 1$ and $g \gg J$, the field sgZ_v detunes site v strongly. Note that H_G is a commuting family of XX bonds, but $[Z_v, X_v X_j] \neq 0$: the controlled Hamiltonian is not commuting. In the frozen-Z_v sector, second-order perturbation theory gives

$$H_{\text{eff}}^{(s)} = H_{G-v} + \frac{J^2}{2sg}\Big(\sum_{j\in N(v)} X_j\Big)^2 + O(J^3/g^2) = H_{G-v} + \frac{J^2}{sg}\sum_{\substack{i<j\\ i,j\in N(v)}} X_i X_j + \text{const.} + O(J^3/g^2), \quad (14)$$

where $N(v)$ are the neighbours of v. The contact acts on two time scales:

- **Fast (rate J):** it removes v. The rest evolves as the graph $G - v$, whose connected components become the entangled clusters.
- **Slow (rate J^2/g):** it couples all neighbours of v to each other. The sign follows the ancilla; with $p_- = \frac{1}{2}$ it is random at each collision and this term dephases.

Check: for the three-qubit chain of [1], contact at the middle site, eigenstate ancilla, $g/J = 5$: Eq. (14) gives $J_{\text{eff}} = 0.2$ and $C_{13} = |\sin 2J_{\text{eff}} t|$, with peaks near $n = 10, 29, 49$ at $J\tau = 0.4$; the paper reports the third peak at $n = 51$.

Predicted structures, Setting I, $g \gg J$.

Graph, contact	Fast: components of $G - v$	Slow, $p_- = 0$: couplings added in $N(v)$	$p_- = \frac{1}{2}$
P_4, end 1	GME triple $\{2,3,4\}$	none (one neighbour)	as fast
P_4, middle 2	Bell pair $(3,4)$; qubit 1 free	1–3 $\to$ triple $\{1,3,4\}$	as fast
Star, hub	nothing	K_3 on leaves $\to$ GHZ-class triple	nothing
Star, leaf 2	GME triple $\{1,3,4\}$	none	as fast
C_4, site 1	GME triple $\{2,3,4\}$	2–4 added	as fast
Paw, vertex 3	Bell pair $(1,2)$; qubit 4 free	clique $\{1,2,4\} \to$ triple	as fast
Paw, triangle vertex 1	GME triple $\{2,3,4\}$	reweights 2–3	as fast
Paw, tail 4	GHZ-class triple $\{1,2,3\}$	none	as fast
Diamond, vertex 2	GME triple $\{1,3,4\}$	1–3 added	as fast
Diamond, vertex 1	GHZ-class triple $\{2,3,4\}$	reweights 2–4	as fast
K_4, site 1	GHZ-class triple $\{2,3,4\}$	reweights all	as fast

These are predictions. Where they fail ($g \sim J$, partial polarization, Setting II) is where new physics is most likely.

5.5 Spectrum of the collision channel

Write the channel (2) as a $4^N \times 4^N$ superoperator and compute its eigenvalues, $\mathcal{E}_v(R_\mu) = \lambda_\mu R_\mu$:

- eigenvalues $\lambda = 1$: stationary operators; their number is the dimension of the fixed-point space;
- other eigenvalues with $|\lambda| = 1$: persistent oscillations;

- the largest $|\lambda| < 1$: the slowest decay, with $1/e$ time $-1/\ln|\lambda|$ collisions.

A plateau over hundreds of collisions is not proof of stationarity. A stationary claim requires a physical density matrix ρ_∞ with trace-norm residual $\|\mathcal{E}_v(\rho_\infty) - \rho_\infty\|_1$ at round-off level, a spectral gap that rules out slower modes, and GME certified directly on the relevant reduced state of ρ_∞. If the fixed-point space has dimension larger than one, ρ_∞ depends on the state before contact; say so.

5.6 Quotient graphs (Setting II)

In the single-excitation sector the exchange Hamiltonian is $\frac{J}{2}$ times the adjacency matrix. If the state is symmetric under $\mathrm{Aut}(G)_v$, vertices in one orbit merge into a single collective mode, and the dynamics runs on a smaller *quotient graph*. Example: a star contacted at the hub reduces to the hub and the symmetric leaf mode, coupled by $\sqrt{3}\,J/2$. Use this to predict oscillation frequencies and which modes a pump can reach.

6 Preliminary results

These pilot calculations fix the starting point. All use $J\tau = 0.4$ and are to be reproduced in Tasks 1–2.

#	Test	Result	Implication
1	Reproduce [1] (3-qubit chain and triangle, $g/J = 20$)	$C_{BC}(n{=}2) = 0.990$ and 0.978; Bell fidelity 0.995 and 0.989	Implementation agrees with published values
2	No-collision baseline, 3-qubit chain and triangle	GMN $= 0.500$ at $Jt = \pi/4$	The XX network alone produces maximal GHZ-class GME
3	Bell routing vs. GME	$\mathrm{GMN}_{ABC} = 3 \times 10^{-4}$ when $C_{BC} = 0.99$	Collisions convert tripartite GME into a Bell pair
4	Ancilla coherence, ZZ contact	$\|+\rangle$ and $\nVdash/2$ give identical network states (10^{-16})	Polarization, not coherence, is the control
5	Coupling-pair table (Sec. 3.3) on P_4	"No" rows: zero negativity on all cuts. (EX, XX) with $\|+\rangle$: largest cut negativity of all pairs (1.45)	Table confirmed; (EX, XX) is an active candidate
6	Long-time limit, Setting I	All entanglement vanishes; many fixed points	GME is transient in Setting I

6.1 Setting I: the contact site selects the components of $G - v$

On a six-qubit path with strong contact, the clusters left entangled are the components of $G - v$, as Eq. (14) predicts at leading order. Fidelities with the corresponding graph states at $n = 2$ (above $\frac{1}{2}$ certifies GME of that subset):

Contact	Components of $G - v$	F, $g/J = 5$	F, $g/J = 20$
Node 1	$\{2, 3, 4, 5, 6\}$	0.927	0.994
Node 3	$\{1, 2\}$ and $\{4, 5, 6\}$	0.928 and 0.927	0.994 and 0.994
Node 6	$\{1, 2, 3, 4, 5\}$	0.927	0.994

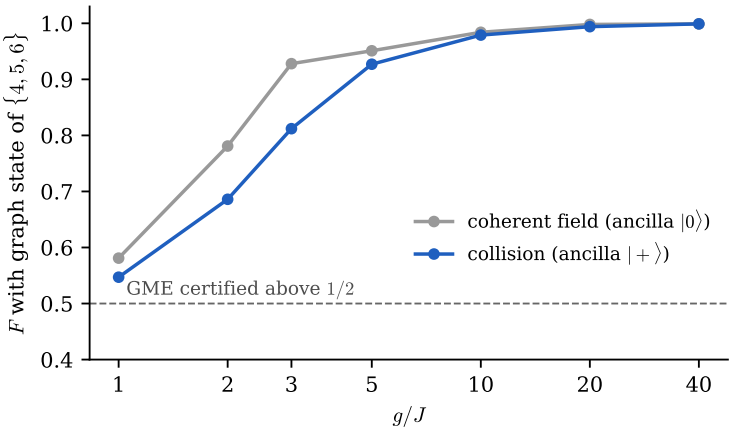


Figure 1: A coherent field ($p_- = 0$) matches or beats the unpolarized collision ($p_- = \frac{1}{2}$) at every coupling strength. Fidelity of sites $\{4, 5, 6\}$ with their graph state; six-site path, contact at node 3, $n = 2$, $J\tau = 0.4$.

In this regime the selection is coherent control by the contact (graph reduction), not a dissipative advantage.

6.2 Setting II: stationary GME stabilized by switching the contact site

Protocol. Exchange star $K_{1,3}$ with exchange contact, pump ancilla $|1\rangle$, $g/J = 5$. Stage 1: n_1 collisions at a leaf. Stage 2: collisions at the hub. After the switch, the three leaves no longer touch the ancilla.

Result. For $n_1 = 4$ the leaf GMN rises from 0.001 before the switch to 0.115 after it.

Exact fixed-point verification. The hub channel was analysed as in Sec. 5.5:

- 9 eigenvalues equal to 1, no other eigenvalue on the unit circle, and next largest $|\lambda| = 0.909$ (1/e time 10.5 collisions). There is no slow mode that could mimic a plateau.
- The fixed point obtained by spectral projection and by 3000-step iteration agree to 2×10^{-7} (limited by the conditioning of the eigenvector matrix). Trace-norm residual $\|\mathcal{E}(\rho_\infty) - \rho_\infty\|_1$: 8×10^{-15} (projection), 4×10^{-249} (iteration).
- The state after 600 hub collisions differs from ρ_∞ by 4×10^{-50} in trace norm.
- GMN of the leaf state of ρ_∞ is 0.1154; two solvers (SCS and Clarabel) agree to 3×10^{-9}. The hub is left exactly pure.

Mechanism. With the contact at the hub, the leaves couple to the rest only through $L^\pm = \sum_i \sigma_i^\pm$, and the leaves' total spin is exactly conserved, $[H_G + g\, h_{a,\mathrm{hub}}, \mathbf{J}_L^2] = 0$. The pump drives each J_L sector to its extremal state, and the sector weights p_{J_L} set in stage 1 are kept. The leaf state of ρ_∞ has rank 2, with eigenvalues 0.678 and 0.322: exactly the weights $p_{1/2}$ and $p_{3/2}$ deposited in stage 1. Consistently,

$$\rho_\infty = \sum_{J_L} p_{J_L}\, |J_L, -J_L\rangle\langle J_L, -J_L| \otimes \rho_{\mathrm{mult}}^{(J_L)}, \qquad p_{J_L} = \mathrm{Tr}[\Pi_{J_L}\, \rho(n_1)]\,. \tag{15}$$

For three leaves the $J_L = \frac{1}{2}$ sector has multiplicity 2, a noiseless subsystem [9]. A hub pump alone ($n_1 = 0$) keeps the state in $J_L = \frac{3}{2}$ and produces nothing.

Correct wording. The fixed-point space is 9-dimensional, so ρ_∞ remembers stage 1. The result is *stabilization* of GME selected by the first contact, not preparation of a unique steady state.

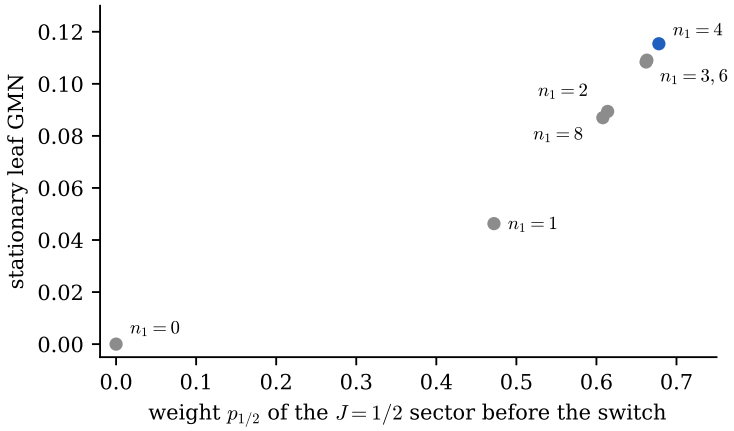


Figure 2: Stationary leaf GMN after the switch rises with the weight that stage 1 deposits in the $J_L = \frac{1}{2}$ sector. $K_{1,3}$, exchange coupling, pump $|1\rangle$, $g/J = 5$, $J\tau = 0.4$.

Controls (pilot, to be reproduced). An unpolarized pump in stage 2 gives GMN $= 10^{-4}$; a hub pump without stage 1 gives 0; a single retained ancilla gives only a transient (peak 0.24, then 0.018 at $n_2 = 600$). Changing one hub–leaf coupling by $0.1J$ keeps the effect. A leaf–leaf coupling or leaf detuning of $0.1J$ removes it within about 10^3 collisions, while $0.01J$ shows no decay within 3×10^3 collisions. These robustness numbers are finite-time observations; Task 6 replaces them by decay rates.

7 Tasks

Work through the tasks in order. Each ends with a gate that must pass before the next task starts.

7.1 The benchmark set

All six connected graphs on four vertices, with their symmetry-inequivalent contact sites (11 in total):

Graph	Edges	$\lvert\mathrm{Aut}(G)\rvert$	Inequivalent contact sites
Path P_4	12, 23, 34	2	end $\{1,4\}$; middle $\{2,3\}$
Star $K_{1,3}$	12, 13, 14	6	hub 1; leaf $\{2,3,4\}$
Cycle C_4	12, 23, 34, 41	8	any
Paw	12, 23, 13, 34	2	degree-3 vertex 3; triangle vertex $\{1,2\}$; tail 4
Diamond $(K_4 - e)$	12, 23, 34, 41, 24	4	degree-3 $\{2,4\}$; degree-2 $\{1,3\}$
Complete K_4	all six	24	any

Parameters.

Axis	Setting I (XX network, ZZ contact)	Setting II (exchange network; EX and XX contacts)
g/J	1, 2, 5, 20	1, 5
Ancilla	$p_- = 0,\ \frac{1}{4},\ \frac{1}{2}$	EX: pump population 1, $\frac{3}{4},\ \frac{1}{2}$; XX: $p_- = 0,\ \frac{1}{2}$
Pre-contact time	$t_0 = 0$ and $t_0 = \pi/4J$	$t_0 = 0$
$J\tau$	0.4 (as in [1]); check at 0.1	0.4
Collisions	$n = 0$–200	$n = 0$–400 and the fixed point

Task 1. Code and validation

1. Implement the channel (2) for any adjacency matrix, coupling family, contact site, η and τ; the protocols (4) and (5); and a single retained ancilla (no reset).
2. Implement partial trace, partial transpose, negativity, concurrence, all 15 partitions, partition exclusion, GMN, fidelity witnesses, QFI, the channel superoperator and graph automorphisms (Appendix A).
3. After each step check $\mathrm{Tr}\rho = 1$, $\rho = \rho^\dagger$ and $\rho \geq 0$ to 10^{-12}. Cross-check the partial-trace update with an explicit Kraus representation.

Gate: every row of the validation table (Sec. 4.5) and the concurrence curves of [1] (Figs. 2–3 and Appendix A) are reproduced.

Task 2. Verify the switching result first

Reproduce Sec. 6.2: the leaf GMN of 0.115, all controls (A-only, B-only, reversed $B \to A$, unpolarized pump, retained ancilla), the channel spectrum, the fixed point by two methods, its residual, and GMN on the leaf state of ρ_∞ with an independent second solver. Scan $n_1 = 0$–12 and $g/J = 1, 2, 5, 10$.
Gate: the result is classified as stationary or metastable from the spectrum and residual, not from a plateau.

Task 3. Coupling pairs and no-collision baselines

1. Run all nine (network, contact) pairs on P_4 for 60 collisions and confirm the table of Sec. 3.3, including the strong activity of (EX, XX).
2. Setting I without collisions, all six graphs, $Jt \in [0, \pi/2]$ (one period). At each time compute all quantities of Sec. 4.

Gate: agreement with the coupling-pair table to machine precision; at $Jt = \pi/4$ the GME type matches the graph-state classes of Sec. 5.3; star and K_4 show $C_{ij} = 0$.

Task 4. Contact-site graph reduction

Setting I, $g/J = 20$, $p_- = 0$ and $\frac{1}{2}$, all 11 sites, $t_0 = 0$ and $t_0 = \pi/4J$. Compare coherent ($p_- = 0$) and unpolarized ($p_- = \frac{1}{2}$) contacts at matched elapsed time and initial state. Test the prediction table of Sec. 5.4: fast clusters at $t \approx \pi/4J$, slow neighbour couplings at $t \sim g/J^2$ for $p_- = 0$, and their absence for $p_- = \frac{1}{2}$.
Gate: every orbit equality of Sec. 5.1 holds; a one-page summary states where leading order works and where it fails.

Task 5. Systematic four-vertex benchmark

Both settings, all 11 sites, the parameters above. For each site, map the reporting label (Sec. 4.10) and the excluded partitions versus g/J and ancilla polarization. In Setting II include fixed points

from the channel spectrum.
Gate: structure maps with validated certificates, and a list of results that none of the tools of Sec. 5 explains.

Task 6. One generalization and synthesis

Choose one direction, not every scan:

- extend the switching result to stars $K_{1,m}$, $m = 3$–6, including the $J_L = 0$ sector for even m, which needs contacts at two or more leaves; derive ρ_∞ and its GMN analytically from p_{J_L};
- or extend graph reduction to the six-site path and a 2×3 grid, including two simultaneous contacts.

In either case, add weak symmetry-breaking perturbations (leaf–leaf coupling ϵ, detuning w) and fit decay rates; test whether they scale as ϵ^2 and w^2. Then label every result by the tool that explains it.
Gate: a general principle, or a clear statement of its limits, and the figures of Sec. 9.

8 Controls and reporting standards

Controls for every headline result.

1. isolated continuation ρ_{free} from the same pre-contact state;
2. coherent-field control (eigenstate ancilla), where the contact allows it;
3. retained-ancilla reference, the continuous protocol of [1];
4. no-switch and reversed-switch sequences for two-stage protocols;
5. a fresh-start or reset comparison whenever the word "transfer" is used.

Wording. Use four distinct terms:

- *redistribution*: a different subset is entangled after the contact than without it;
- *generation*: entanglement appears where none existed before;
- *stabilization*: a certified state is a fixed point of the channel (Sec. 5.5);
- *relocation* or *transport*: entanglement initially in one subset is moved to another, which requires the reset and reference-qubit controls.

What to report. For each headline result: the graph and node labels, the exact ancilla density matrix, the coupling normalization, $J\tau$, g/J, t_0, collision counts, the subset, the solver and its residuals, and whether the result is transient, metastable or stationary. Store raw density matrices at the sampled steps.

Figures of merit. Fix before running:

- peak certified value in $[0, n_{\max}]$ and its collision index;
- time-integrated certified signal, $\sum_n \mathrm{GMN}_S(n)\,\tau$;
- fraction of steps with $\mathrm{GMN}_S > \epsilon_{\text{cert}}$;
- the fixed-point value, when a fixed point exists.

Pitfalls.

- The free graph-state time $t = \pi/4J$ falls between $n = 1$ and 2 for $J\tau = 0.4$. State the time grid in every figure.
- Use graph symmetries only when the contact site, the initial state and η respect them.
- A small nonzero GMN may be solver noise; apply Sec. 4.7.

9 Expected outputs and result hierarchy

Figures.

1. Graph gallery: contact-site orbits, no-collision cluster map and contact-controlled cluster map.
2. Structure timelines at matched times: rows for the 6 pairs, 4 triples and the full state, shaded where certified, with certified depth lower bounds.
3. Strong-contact comparison: graph reduction and the J^2/g slow scale, with coherent and unpolarized contributions separated.
4. Switching panel: GMN time series across the switch, channel spectrum and fixed-point residual, conserved J_L sector weights, and decay under symmetry breaking.

Result hierarchy.

- If stabilization of GME by contact switching holds beyond $K_{1,3}$, it is the central result; graph reduction is a complementary topology-control mechanism.
- If it holds only for long finite times under realistic perturbations, present it as controllable metastable multipartite localization, with lifetimes.
- If neither holds generally, the main result is topology-predictive coherent restructuring of entanglement, positioned carefully against Zeno and graph-state protocols [6–8].

No phase-transition or scaling claim is needed.

10 Outlook

Noise and load sites. Add a fixed local noise channel at one site and designate one or two "load" sites where entanglement is needed. Ask which depth survives best there, and whether the contact site or a schedule of contact sites can steer it there. In the single-excitation sector of exchange networks, a lossy load site acts as a trap, as in noise-assisted transport [18, 19]; here the target is entanglement depth rather than transport efficiency. For static states, GHZ-type entanglement is more fragile than cluster- or W-type under local noise [20]. The open question is dynamical: can collisions regenerate or re-route depth faster than noise removes it?

Larger networks. There are 21 connected graphs on five vertices and 112 on six. The symmetry, fast/slow and spectral tools scale with graph size; numerical cost scales as 4^N. Use the tools to choose a short list.

Relocation. To show that entanglement moves rather than re-forms: compare with a fresh start; entangle a reference qubit with the source subset and test its entanglement with the target; and

vary the path length. Whether Setting I or II is better suited must be decided from these tests, not from commutation arguments.

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A Code skeletons

Python with NumPy, SciPy and CVXPY. Qubit 0 is the leftmost tensor factor. In the collision routines the ancilla is qubit 0 and network vertex k is qubit $k+1$.

Operators, partial trace, partial transpose

```
import numpy as np, itertools, cvxpy as cp
from scipy.linalg import expm

I2 = np.eye(2)
X = np.array([[0, 1], [1, 0]], complex)
Y = np.array([[0, -1j], [1j, 0]])
Z = np.diag([1., -1.]).astype(complex)
SP = np.array([[0, 1], [0, 0]], complex)   # sigma^+ = (X + iY)/2
SM = SP.T.copy()

def op(o, i, n):
    """Single-qubit operator o on qubit i of n."""
    out = np.array([[1.]], complex)
    for k in range(n):
        out = np.kron(out, o if k == i else I2)
    return out

def ptrace(rho, keep, n):
    """Reduced state on the qubits in `keep` (sorted list)."""
    t = rho.reshape([2] * 2 * n)
    gone = [k for k in range(n) if k not in keep]
    for c, k in enumerate(sorted(gone, reverse=True)):
        m = n - c                          # qubits still present
        t = np.trace(t, axis1=k, axis2=k + m)
    d = 2 ** len(keep)
    return t.reshape(d, d)

def ptranspose(rho, A, n):
    """Partial transpose on the qubits in A."""
    t = rho.reshape([2] * 2 * n)
    for s in A:
        t = np.swapaxes(t, s, s + n)
    return t.reshape(2 ** n, 2 ** n)

def tracenorm(M):
    return np.abs(np.linalg.eigvalsh((M + M.conj().T) / 2)).sum()
```

Negativity, concurrence, cuts and partitions

```
def negativity(rho, A, n):
    ev = np.linalg.eigvalsh(ptranspose(rho, A, n))
    return (np.abs(ev).sum() - 1) / 2

def psd_sqrt(M):
    w, V = np.linalg.eigh((M + M.conj().T) / 2)
    return V @ np.diag(np.sqrt(np.clip(w, 0, None))) @ V.conj().T

def concurrence(r, tol=1e-12):
    yy = np.kron(Y, Y)
    rt = yy @ r.conj() @ yy
    s = psd_sqrt(r)
    mu = np.linalg.eigvalsh(s @ rt @ s)          # Hermitian, should be >= 0
    if mu.min() < -1e3 * tol:
        raise ValueError('negative eigenvalue %.2e: check rho' % mu.min())
    lam = np.sort(np.sqrt(np.clip(mu, 0, None)))[::-1]
    return max(0., lam[0] - lam[1] - lam[2] - lam[3])
```

```
def cuts(n):
    """All bipartitions A|B, each listed once."""
    out = []
    for k in range(1, n // 2 + 1):
        for A in itertools.combinations(range(n), k):
            if 2 * k == n and 0 not in A:
                continue
            out.append(list(A))
    return out                              # 2**(n-1) - 1 cuts

def set_partitions(s):
    """All partitions of the list s into non-empty blocks."""
    if len(s) == 1:
        yield [s]; return
    first, rest = s[0], s[1:]
    for p in set_partitions(rest):
        for i in range(len(p)):
            yield p[:i] + [[first] + p[i]] + p[i + 1:]
        yield [[first]] + p                 # 15 partitions for 4 qubits

def partition_excluded(rho, P, n, tol=1e-9):
    """True if some cut that does not split a block of P is NPT."""
    for r in range(1, len(P)):
        for blocks in itertools.combinations(P, r):
            A = [q for b in blocks for q in b]
            if negativity(rho, A, n) > tol:
                return True
    return False
```

GMN by semidefinite programming

```
def gmn(rho, n, delta=1e-9):
    """Returns (value, status). Uses GMN <= min_M N_M to skip the SDP."""
    cs = cuts(n)
    nmin = min(negativity(rho, A, n) for A in cs)
    if nmin < delta:
        return 0.0, 'GMN < %.0e (bound)' % delta
    d = 2 ** n
    W = cp.Variable((d, d), hermitian=True)
    cons = []
    for A in cs:
        P = cp.Variable((d, d), hermitian=True)
        Q = cp.Variable((d, d), hermitian=True)
        QT = Q
        for s in A:
            QT = cp.partial_transpose(QT, [2] * n, s)
        cons += [W == P + QT, P >> 0, np.eye(d) - P >> 0,
                 Q >> 0, np.eye(d) - Q >> 0]
    prob = cp.Problem(cp.Minimize(cp.real(cp.trace(W @ rho))), cons)
    prob.solve(solver='CLARABEL')
    return max(0.0, -prob.value), prob.status
```

Fidelity-witness bound and QFI

```
def alpha_bisep(psi, n):
```

```
    """Largest squared Schmidt coefficient over all cuts."""
    r = np.outer(psi, psi.conj())
    return max(np.linalg.eigvalsh(ptrace(r, A, n))[-1] for A in cuts(n))

def qfi(rho, Jop):
    lam, V = np.linalg.eigh(rho)
    Jm = V.conj().T @ Jop @ V
    F = 0.0
    for i in range(len(lam)):
        for j in range(len(lam)):
            s = lam[i] + lam[j]
            if s > 1e-12:
                F += 2 * (lam[i] - lam[j]) ** 2 / s * abs(Jm[i, j]) ** 2
    return F
```

Collision step

```
def h(kind, i, j, n):
    if kind == 'xx': return op(X, i, n) @ op(X, j, n)
    if kind == 'zz': return op(Z, i, n) @ op(Z, j, n)
    if kind == 'ex': return 0.5 * (op(SP, i, n) @ op(SM, j, n) + op(SM, i, n) @ op(SP,
    j, n))

def collision_unitary(edges, N, v, J, g, tau, net='xx', con='zz'):
    n = N + 1                                  # ancilla = qubit 0
    H = sum(J * h(net, i + 1, j + 1, n) for i, j in edges) + g * h(con, 0, v + 1, n)
    return expm(-1j * H * tau)

def collide(U, eta, rho, N):
    big = U @ np.kron(eta, rho) @ U.conj().T
    return ptrace(big, list(range(1, N + 1)), N + 1)
```

Channel spectrum and fixed point

```
def superoperator(U, eta, N):
    d = 2 ** N
    S = np.zeros((d * d, d * d), complex)
    for k in range(d * d):
        B = np.zeros(d * d, complex); B[k] = 1
        S[:, k] = collide(U, eta, B.reshape(d, d), N).reshape(-1)
    return S

def spectrum_summary(S, tol=1e-10):
    ev = np.linalg.eigvals(S); a = np.abs(ev)
    n_one = int(np.sum(np.abs(ev - 1) < tol))          # fixed-point dimension
    n_osc = int(np.sum((a > 1 - tol) & (np.abs(ev - 1) > tol)))
    lam2 = a[a < 1 - tol].max()                         # slowest decay
    return n_one, n_osc, lam2, -1 / np.log(lam2)

def fixed_point(S, rho, n_iter=3000):
    d = rho.shape[0]; x = rho.reshape(-1)
    for _ in range(n_iter):
        x = S @ x
    r = x.reshape(d, d)
    res = tracenorm((S @ x).reshape(d, d) - r)
    return r, res
```

Example: star switching (Task 2)

```
star = [(0, 1), (0, 2), (0, 3)]            # hub = vertex 0
pump = np.diag([0, 1]).astype(complex)     # ancilla |1>
UL = collision_unitary(star, 4, v=1, J=1., g=5., tau=0.4, net='ex', con='ex')
UH = collision_unitary(star, 4, v=0, J=1., g=5., tau=0.4, net='ex', con='ex')
rho = np.zeros((16, 16), complex); rho[0, 0] = 1
for _ in range(4):                          # stage 1: leaf
    rho = collide(UL, pump, rho, 4)
S = superoperator(UH, pump, 4)
print(spectrum_summary(S))                  # (9, 0, 0.909..., 10.5...)
rinf, res = fixed_point(S, rho)
print(res, gmn(ptrace(rinf, [1, 2, 3], 4), 3))   # ~1e-249, 0.1154
# Clarabel may report 'optimal_inaccurate' here: confirm with
# solver='SCS', eps=1e-9 (gives 0.1154, agreeing to 3e-9).
```